Treatment, Management, and Prevention

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Learning Objectives

Upon completion of this section, you will be able to

- Describe general management goals for patients with asthma,
- Describe modifying factors that might affect how environmental triggers cause/exacerbate asthma, and
- Identify at least three things you can advise patients to do to decrease exposure to allergens and irritants.

Treatment and Management Overview

This Case Study discusses the role environmental factors play in

- **Causing**
- **Triggering, and**
- **Exacerbating asthma.**

This Case Study does not comprehensively review asthma treatment and management.

The treatment and management of environmental asthma follow the guidelines set forth by the National Heart, Lung, and Blood Institute, with special emphasis on the management of the patient's environment [NHLBI 2007].

Pharmaceutical intervention forms the basis of asthma treatment. Asthma medications are generally categorized as

- Relievers: medications used on an as-needed basis that act quickly to reverse bronchoconstriction and relieve symptoms or
- Controllers: medications taken daily on a long-term basis to keep asthma under clinical control mainly through their anti-inflammatory effects.

A stepwise approach is taken for the long-term management of asthma after confirming the diagnosis and assessing the severity of disease [NHLBI 2007, GINA 2011].

Goals for the general management of a patient with asthma include

- Preventing chronic asthma symptoms and exacerbations (day and night),
- Maintaining the patient's "normal" activity (including exercise and other physical activities),
- Regaining and maintaining normal or near-normal lung function, and
- Prescribing optimal pharmacotherapy with minimal or no adverse effects.

Management includes careful monitoring of the patient's response to treatment and appropriate adjustments. It also includes educating the patient and family regarding primary and secondary preventive measures [NHLBI 2007; GINA 2011].

Predisposing Factors

Atopy, the genetic predisposition for the development of an immunoglobulin E (IgE)-mediated response to common airborne allergens, is the strongest identifiable predisposing factor for developing asthma [NHLBI 1997; Holla et al. 2002; NHLBI 2007; Busse and Rosenwasser 2003].

Most children with asthma have allergic rhinitis, a major independent risk factor for asthma. Rhinitis and asthma can be viewed as manifestations of one syndrome – the chronic allergic respiratory syndrome – in different parts of the respiratory tract [Togias 2003].

Certain immune system components, such as the T-helper phenotype, are determined in the first year of life by environmental exposure to respiratory infections or environmental allergens in genetically predisposed individuals [Robinson et al. 2004; Luft et al. 2004; Larche et al. 2003; Umetsu et al. 2003].

Exposure to Allergens and Risk of Asthma

Studies of exposure to allergens and risk of asthma have yielded paradoxical results.

- Exposure to some pets appears to increase the risk of asthma and wheezing in older children, yet lower the risk among young children [Apelberg et al. 2001].
- House dust mite and cockroach allergens appear to have a positive linear relationship, whereas cat allergens appear to act quite differently, with maximum sensitization developing at moderate exposure levels.
- Very low levels of cat allergen exposure are likely to induce no response; very high levels are likely to develop a form of tolerance [Murray et al. 2001].
- Decreased exposure to infections and allergens in early childhood has been linked to the increased incidence of asthma in industrialized countries (the “hygiene hypothesis”) [Liu and Murphy 2003].

Hygiene Hypothesis

The hygiene hypothesis of asthma states that naturally occurring infections and allergen exposures might essentially immunize against the development of asthma and allergic and autoimmune diseases. The modern emphasis on cleanliness or “sanitizing the environment” may have reduced this natural immunotherapy over the past century and might be a factor in the global increase of these conditions [Liu and Murphy 2003].The differences in health outcomes from exposure are due to important moderating variables, such as

- Age of exposure,
- Timing of exposure relative to disease development,
- Dose and frequency of exposure,
- Co-exposures, and
- Genetic predispositions in response [Song and Liu 2003].

Growing up on a farm may protect against developing asthma and allergic rhinoconjunctivitis [Von Essen 2001]. An important study in 2002 showed that exposure of young children to older children at home or to other children in childcare settings protects against the development of asthma and frequent wheezing later in childhood [Ball et al. 2000].

Primary Prevention Strategies in Children

Well-documented primary prevention strategies for asthma.

Avoid smoking and environmental tobacco smoke (ETS).

For children, studies indicate that in utero exposure to tobacco smoke products is an important predictor of wheezing within the first year of life. Exposure to ETS places children at increased risk for the development and exacerbation of asthma as well as

- Sinusitis,
- Otitis media,
- Bronchiolitis, and
- Diminished pulmonary function.

Both in utero and passive (environmental) tobacco smoke exposure adversely affect pulmonary function, and predispose to asthma symptoms and possibly bronchial hyper responsiveness in childhood. Exposure to tobacco smoke products in utero is a risk factor for wheezing in the first year of life [Tager et al. 1993]. Children who have asthma and whose parents smoke have more frequent asthma attacks and more severe symptoms [Weitzman et al. 1990; Martinez et al. 1992].

Avoid exposure to insect allergens.

House dust mite and cockroach allergens have a very close association between exposure and the sensitization of an individual [Murray et al. 2001].

Avoid exposure to molds.

Exposure to mold in homes as much as doubles the risk of asthma development in children [Jaakkola et al. 2005].

Breast-feed infants.

A study demonstrated that exclusively breastfeeding for the first 4 months is associated with a statistically significant decrease in the risk of asthma and wheezing in children until the age of 6 years [Dell and To 2001].

Primary Prevention in Adults

In adult-onset asthma, primary prevention relies mainly on smoking cessation and control of workplace exposures. Studies of factory workforces in the past decade have provided consistent evidence of exposure-response relationships for both sensitization (IgE production) and asthma [Taylor 2001; Jeebhay et al. 2001].

New-onset occupational asthma may be immunological or nonimmunological in origin. The immunologic variants are usually caused by high molecular-weight allergens such as grain dust and animal or fish protein. Symptoms may take months or years to develop.

A brief, high-level exposure to a strong irritant can precipitate nonimmunologic occupational asthma. Symptoms occur immediately or within a few hours of the exposure. Multiple lower level exposures to an irritant can also cause asthma. The worker should be removed from further exposure once the diagnosis of occupational asthma is established, whether immunologic or nonimmunologic in origin. Continued exposure to sensitizers or irritants following sensitization may cause persistent problems that can lead to permanent impairment. In addition, once sensitized, individuals may have a substantial response to extremely low levels of sensitizers or irritants. If the diagnosis is made in a timely fashion and steps are taken to stop exposure, most workers experience improvement. Prevention is the best therapeutic intervention [Bardana 2003].

Avoidance of exposure to occupational irritants and allergens is the mainstay of primary prevention. Especially notorious for producing occupational asthma are jobs that use

- Isocyanates,
- Enzymes, or
- Latex.

Prospective surveillance can detect the development of specific IgE antibody before the onset of allergic symptoms. This allows continuing interventions to reduce exposures and minimize or eliminate those associated with symptoms. Workers with IgE to specific allergens can continue to work in the industry symptom-free for their entire careers. This indicates that exposures needed to induce sensitization are different and probably lower than exposures needed to elicit allergic symptoms [Wisnewski et al. 2006; Sarlo and Kirchner 2002].

Secondary Prevention in Children and Adults

Patients can take a number of steps to reduce or avoid exposure to:

- Pollutants,
- Irritants, and
- Allergens

that may trigger or exacerbate asthma episodes [Williams et al. 2003; AAPCEH 2003]. The National Environmental Education and Training Foundation outlined possible preventive measures in Environmental Management of Pediatric Asthma: Guidelines for Health Care Providers. Summarized below are those environmental intervention guidelines [NEETF 2005]. It is important to note that no single intervention will likely achieve sufficient benefits to be cost effective and that a comprehensive environmental intervention may be needed to improve asthma-associated morbidity [GINA 2011].

Dust Mites

No matter how clean the home is, dust mites cannot be eliminated. However, household interventions can decrease exposure to dust mites and possibly reduce asthma exacerbations [Ehnert et al. 1992]. Cleaning with a high-efficiency particulate air (HEPA) filter vacuum is particularly effective in removing allergens and thus decreasing asthma symptoms [McDonald et al. 2002; Platts-Mills et al. 2001].

Listed below are recommended steps to reduce dust mites in the home [NEETF 2005].

- Remove carpet from bedrooms.
- Use an air conditioner or dehumidifier to reduce household humidity.
- Remove upholstered furniture.
- Replace draperies with blinds or other wipeable window coverings.
- Encase pillow and mattress in allergen impermeable cover.
- Remove humidifiers.
- Replace wool or feathered bedding with synthetic materials that will withstand repeated hot water washing.
- Use a damp mop or rag to remove dust (a dry cloth just stirs up mite allergens).
- Vacuum regularly using a cleaner with a HEPA filter or a double-layered microfilter bag (try not to vacuum when the asthmatic is in the room).
- Wash and thoroughly dry stuffed toys weekly in hot water, or freeze them weekly.
- Wash bedding in hot water (at least 130ºF) weekly.

Animal Allergens

Modifications to the home environment can significantly reduce animal allergens and the frequency of asthma episodes [Williams et al. 2003]. The following steps can reduce exposure to animal allergens.

- Find a new home for indoor cats, dogs, and pet rodents that have caused allergy symptoms.
- Keep pets outside.
- Select low-dander pets in place of those with fur or feathers.

If those options are not possible, the following steps **may** help reduce exposure.

- Keep pets out of the bedroom.
- Enclose mattresses and pillows in zippered plastic cases.
- Remove carpets.
- Vacuum regularly using a cleaner with a HEPA filter or a double-layered microfilter bag (try not to vacuum when the asthmatic is in the room).
- Use a portable air cleaner with HEPA filter for the child’s bedroom.
- Keep pets off furniture.

Cockroach Allergen

The first step in limiting cockroach allergens is to keep the house clean and in good shape [O'Connor and Gold 1999]. In general, use the least hazardous methods of roach control first.

Food

- Clean up all food items and crumbs.
- Limit spread of food around house, especially bedrooms.
- Restrict food consumption to the kitchen and dining room.
- Store food (including pet food) in closed containers.

Hygiene and maintenance

- Fix water leaks under sinks.
- Mop the kitchen floor and clean countertops at least once a week.
- Check for and plug crevices outside your house that cockroaches may enter.
- Caulk or patch holes in walls, cupboards, and cabinets.

Pest management

- Use the integrated pest management (IPM) approach for least toxic extermination methods first.
- Use boric acid powder under stoves and other appliances.
- Use bait stations and gels.
- Use outdoor treatments as much as possible to prevent insects from entering your house.
- If those steps are unsuccessful, seek help from a professional, licensed exterminator rather than spraying chemicals yourself.
- Stay away from the house for several hours after pesticides are applied.
- Avoid using liquid sprays inside the house, especially near places children crawl, play, or sleep.

Mold and Mildew

Mold spores are allergens found indoors and outdoors. Outdoor molds are present year-round throughout the West (lower altitudes) and South, and in the North during the fall. Outdoor molds in the North generally peak in late summer. There is no definite seasonal pattern to molds that grow indoors. Moisture control is the key step in limiting indoor mold growth [Krieger and Higgins 2002].

Tips to help keep exposure to mold spores as low as possible.

- Use air-conditioning to cool the house; evaporative coolers are not recommended.
- When first turning on home or car air-conditioners, leave the room or drive with the windows open for several minutes to allow mold spores to disperse.
- Use a dehumidifier or air-conditioner (non-evaporative or water-filled type) to maintain relative humidity below 50%.
- Do not use a humidifier.
- Check faucets, pipes, and ductwork and repair any that are leaking.
- Clean mold with chlorine solution diluted 1:10 with water.
- Do not install carpet and wallpaper in rooms prone to dampness.
- Leave a light on inside a closet that has mold in it to dry the air.
- Install and use exhaust fans in the kitchen, bathrooms, and damp areas.
- Vent bathrooms and clothes dryers to the outside.
- Remove decaying debris from the yard, roof, and gutters.
- Avoid raking leaves, mowing lawns, or working with peat, mulch, hay, or dead wood if you are allergic to mold spores.

Environmental Tobacco Smoke

Cigarette smoke contains many toxic chemicals and irritants. Approximately 42% of children 2 months to 11 years of age live in a home with at least one smoker [Pirkle et al. 1996]. Children exposed to tobacco smoke have increased asthma exacerbations. Studies suggest that asthma symptoms may be less severe for asthmatic children if parents expose them to less cigarette smoke [Murray and Morrison 1993]. Complete cessation of indoor smoking in the homes of children with asthma may be needed to achieve significant health improvement [Lodrup and Carlsen 2001]. The following are the most important preventive strategies to reduce exposure to environmental tobacco smoke.

- Keep your home and car smoke-free.
- If you smoke, do not smoke near children or other nonsmokers.
- Seek support to quit smoking; consider aids such as nicotine gum, patch, and medication from your doctor to help you in quitting.
- Change clothes after smoking while you are in the process of cutting down on the number of cigarettes.
- Choose smoke-free childcare and social settings.
- Seek smoke-free environments in restaurants, theaters, and hotel rooms.

Indoor Air Pollution

For **indoor air pollution**, the two **best approaches to reducing indoor air pollution** are **source control** and **ventilation**. Listed below are specific steps for improving indoor air quality.

- Limit use of products and materials that emit strong odors and irritants, such as
  - Air freshener sprays,
  - Chalk dust,
  - Cleaning products,
  - Hair sprays,
  - Insect sprays.
  - Paint fumes,
  - Sawdust,
  - Smoke,
  - Strong perfumes, and
  - Talcum powder.
- Moderate indoor humidity and moisture (relative humidity between 35-50%).
- Use good housekeeping practices to reduce the presence of airborne particles.
- Install an exhaust fan close to the source of airborne contaminants or odors and vent it to the outside.
- Properly ventilate the room in which a fuel-burning appliance is used.
- Ensure that the doors of wood-burning stoves fit tightly.
- Follow manufacturer's instructions when using an unvented kerosene or gas space heater.
- Ensure fireplaces are properly vented so smoke escapes through the chimney.
- Never use a gas-cooking appliance as a heating source.
- Open windows, especially when pollutant sources are in use (this option must be balanced against the concern of mold allergy or other plant allergens and outdoor air pollution).

Outdoor Air Pollution

Outdoor air pollution, especially ozone and particulate matter, can increase asthma symptoms. Ways to limit exposure to outdoor air pollution.

- Monitor air quality and pollen levels and keep children indoors when pollutants are high.
- Avoid sustained contact with vehicle exhaust emissions and diesel fumes (such as student exposure to idling school buses).
- Use High-Efficiency Particulate Air (HEPA) filters in household vents.
- If possible, move to a less polluted location.
- Schedule outdoor activities for times when ozone levels are lowest, typically in the morning.

Desensitization

For some cases, consider desensitization – especially if environmental control fails to decrease asthma exacerbations.

- Specific immunotherapy involves the administration of allergen extracts to achieve clinical tolerance of the allergens that cause symptoms in patients with allergic conditions.
- Immunotherapy can be effective in patients with mild forms of allergic disease, and in those who do not respond well to standard drug therapy.
- Effects of specific immunotherapy take longer to manifest, but once established, specific immunotherapy may give long-lasting relief of allergic symptoms, whereas the benefits of drugs only last as long as they are continued [Frew 2003b; Nelson 2003].

Key Points

- Every practitioner who treats asthma patients should have general goals for management.
- Environmental triggers can cause or exacerbate asthma.
- Patients can take a number of steps to reduce or avoid exposure to the pollutants, irritants, and allergens that may trigger or exacerbate asthma episodes.

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